

# Biostatistics — Unit 1

## Concept of Biostatistics

### 1. What is Biostatistics?

Biostatistics is the application of statistical methods to biological, medical, public health, and health-related research. It bridges the gap between raw biological data and meaningful scientific conclusions. The word is derived from Greek: *bios* (life) + *statistike* (the science of state data).

**Key Definition:**

**Biostatistics** = The science of collecting, organizing, summarizing, analyzing, and interpreting numerical data obtained from biological experiments and health-related studies.

### 2. Historical Development

The development of biostatistics traces through key milestones:

| Period | Contributor    | Contribution                                 |
|--------|----------------|----------------------------------------------|
| 1662   | John Graunt    | Bills of Mortality – first life tables       |
| 1700s  | Thomas Bayes   | Bayesian probability theorem                 |
| 1800s  | Francis Galton | Regression and correlation concepts          |
| 1900s  | Karl Pearson   | Chi-square test, standard deviation          |
| 1920s  | R.A. Fisher    | ANOVA, experimental design, F-test           |
| Modern | Many scholars  | Survival analysis, clinical trials, genomics |

### 3. Scope and Importance

Biostatistics is essential in every stage of medical and biological research — from designing a study to interpreting its results and drawing valid conclusions.

| Area            | Application Example                                              |
|-----------------|------------------------------------------------------------------|
| Clinical Trials | Testing whether Drug A reduces blood pressure better than Drug B |
| Epidemiology    | Calculating disease incidence and prevalence rates               |
| Genetics        | Analyzing gene frequency in a population                         |
| Public Health   | Studying the effect of vaccination on infection rates            |
| Agriculture     | Comparing crop yields under different fertilizers                |
| Pharmacology    | Dose-response relationship analysis                              |

## 4. Types of Data in Biostatistics

Before applying any statistical technique, it is critical to understand what kind of data you are working with.

### 4.1 Qualitative (Categorical) Data

Data that represents categories or labels, with no inherent numerical value.

Examples: Blood group (A, B, AB, O), Gender (Male/Female), Disease status (Infected/Not Infected)

### 4.2 Quantitative (Numerical) Data

Data that represents measurable quantities.

| Sub-type   | Description             | Example                           |
|------------|-------------------------|-----------------------------------|
| Discrete   | Countable whole numbers | No. of patients: 0, 1, 2, 3...    |
| Continuous | Any value in a range    | Weight: 65.4 kg, Height: 172.3 cm |

### 4.3 Scales of Measurement

| Scale    | Properties                    | Example                              |
|----------|-------------------------------|--------------------------------------|
| Nominal  | Categories, no order          | Blood type, Eye color                |
| Ordinal  | Categories with order         | Pain scale: Mild < Moderate < Severe |
| Interval | Equal intervals, no true zero | Temperature in °C                    |
| Ratio    | Equal intervals, true zero    | Height, Weight, Blood pressure       |

## 5. Numerical Example — Calculating Basic Statistics

A health researcher recorded the Systolic Blood Pressure (mmHg) of 10 patients:

Data: 120, 135, 128, 142, 118, 130, 125, 138, 122, 140

### Step 1 — Count (n):

n = 10 patients

### Step 2 — Sum:

Sum = 120+135+128+142+118+130+125+138+122+140 = 1,298

### Step 3 — Mean (Average):

Mean = Sum / n = 1298 / 10 = 129.8 mmHg

### Interpretation:

The average blood pressure of the group is 129.8 mmHg, which is in the 'elevated' range (Normal: <120 mmHg, Elevated: 120-129 mmHg, Stage 1 Hypertension: 130-139 mmHg). This kind of insight is the core purpose of Biostatistics: turning raw numbers into health decisions.

## 6. Branches of Biostatistics

| Branch                     | Focus                                                 |
|----------------------------|-------------------------------------------------------|
| Descriptive Biostatistics  | Summarizing and describing data (mean, SD, graphs)    |
| Inferential Biostatistics  | Drawing conclusions about populations from samples    |
| Vital Statistics           | Analysis of birth, death, marriage, and disease rates |
| Demography                 | Study of human population size, structure, and change |
| Epidemiological Statistics | Distribution and determinants of diseases             |
| Clinical Biostatistics     | Design and analysis of clinical trials                |

## 7. Population vs. Sample

A fundamental concept: researchers rarely study an entire population. Instead, they study a sample and infer conclusions.

| Concept    | Definition                        | Example                                  |
|------------|-----------------------------------|------------------------------------------|
| Population | The entire group of interest      | All diabetic patients in India           |
| Sample     | A subset drawn from population    | 500 diabetic patients from 5 hospitals   |
| Parameter  | Numerical measure of a population | True mean glucose level of all diabetics |
| Statistic  | Numerical measure of a sample     | Sample mean glucose level (500 patients) |

The science of Biostatistics equips researchers with the tools to make valid, evidence-based conclusions — forming the backbone of modern medicine, public health, and biological research.

## 8. Key Formulae at a Glance

| Formula            | Expression                          | Use                     |
|--------------------|-------------------------------------|-------------------------|
| Mean               | $\bar{x} = \sum x / n$              | Average value           |
| Sample Size        | $n = (Z^2 \times p \times q) / E^2$ | Study design            |
| Relative Frequency | $f/n \times 100$                    | Percentage distribution |
| Prevalence Rate    | Cases / Population $\times K$       | Disease burden          |
| Incidence Rate     | New cases / Person-years $\times K$ | Disease occurrence      |

This PDF covers the foundational concept of Biostatistics. Subsequent PDFs cover each analytical technique in detail.